

Math 341

2026-2027 Autumn – Syllabus

Lectures: Monday : 13:40-14:30 M04

Friday: 9:40-11:30 M104

	Tentative Outline of The Course	
Week	Topics	Section
1.	Graphs as Models Representation of Graphs Graph Parameters Common Graph Classes	1.1 1.2 1.3 1.4
2.	Graph Operations Distance Bipartite Graphs	1.5 1.6 1.7
3.	Trees Tree Algorithms	2.1 2.2
4.	Connectivity	2.3
5.	Menger's Theorem	2.4
6.	Eulerian Graphs Graph Isomorphisms	3.1 3.2
7.	Degree Sequences Degeneracy	3.3 3.4
8.	Applications of Colourings Colouring Bounds	4.1 4.2
9.	Colouring and Operations Extremal k-chromatic Graphs	4.3 4.4
10.	The Four Colour Theorem	5.1
11.	Planar Graphs Kuratowski's Theorem	5.2 5.3
12.	Dual Graphs and Geometry	5.4
13.	Finding Hamiltonian Cycles Hamiltonian Planar Graphs	6.1 6.3
14.	Bipartite Matchings	7.1

Exams and Grading:

(30 pts) Exam 1: November 18, 2026 at 17:40

(30 pts) Exam 2: December 16, 2026 at 17:40

(40 pts) Final Exam: TBA

A make-up exam will be given after the final exam for those who missed one exam for a valid reason. Students who miss all the exams will be given the grade NA.

Textbook: Allan Bickle, Fundamentals of Graph Theory